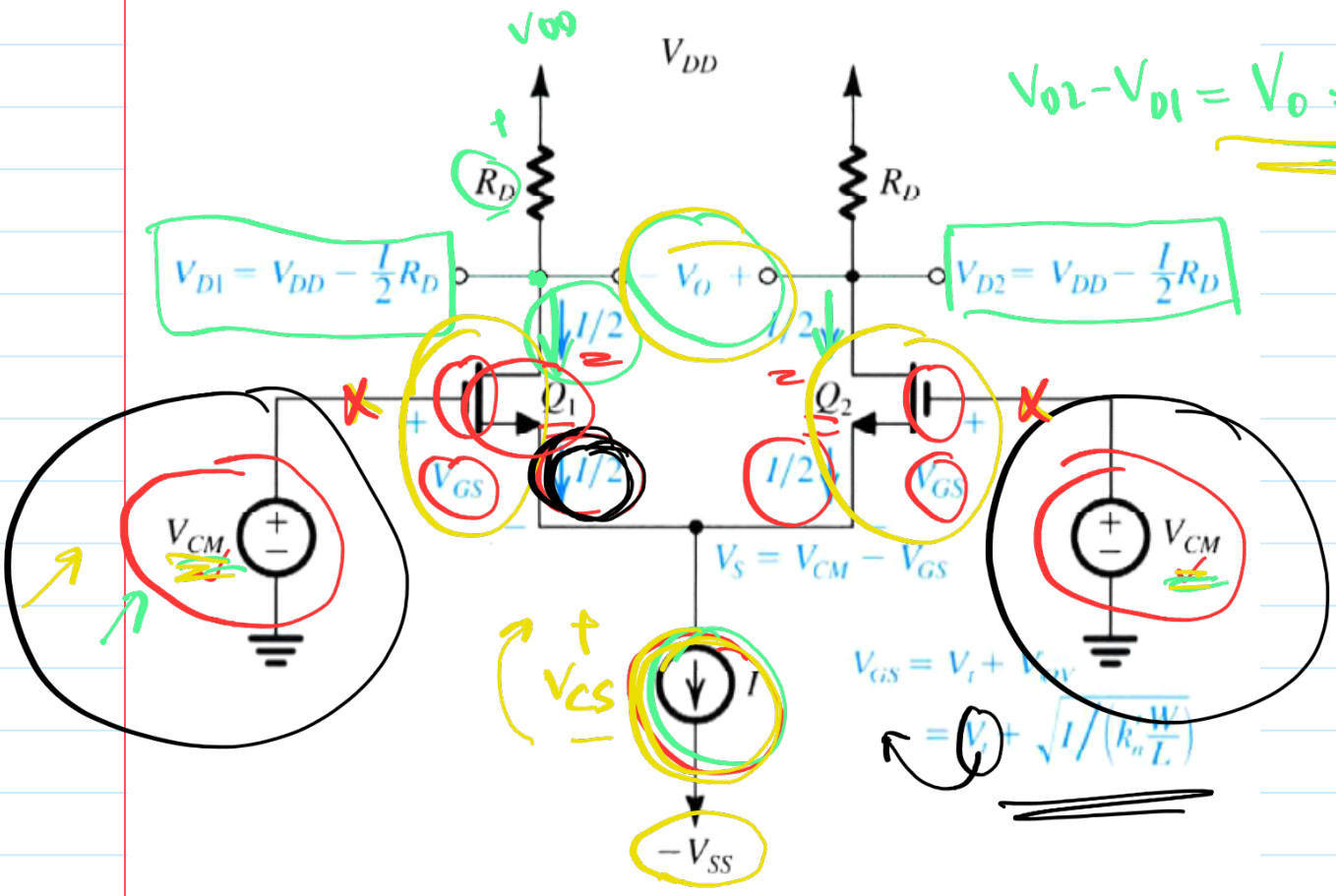
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 9.1 The MOS Differential Pair

Recap



$V_{O2} - V_{O1} = V_O = 0$

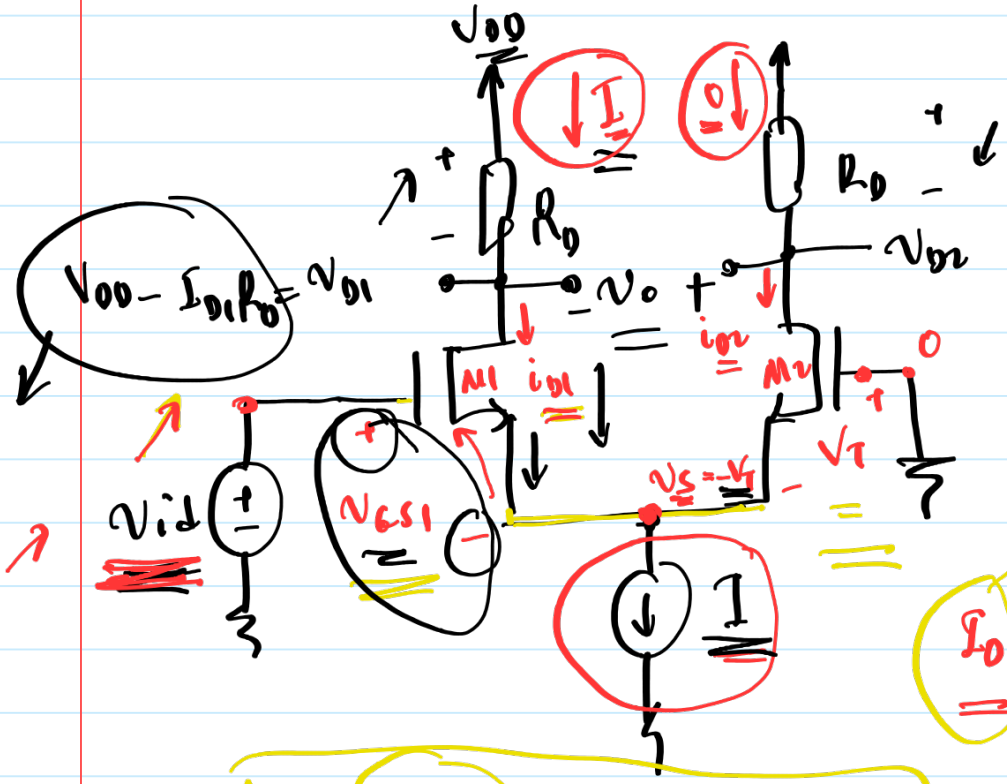
$$V_{CMmax} = V_t + V_{D1,2} = V_t + V_{DD} - \frac{I}{2}R_D$$

$$V_{CMmin} = -V_{SS} + V_{CS} + V_t + V_{OV}$$

$V_{CMmin} \leq V_{CM} \leq V_{CMmax}$

$$I_D = \frac{I}{2} = \frac{1}{2} k_n' \frac{W}{L} V_{ov}^2 \Rightarrow \boxed{V_{ov} = \sqrt{\frac{I}{k_n' \frac{W}{L}}}}$$

Operation with a Differential Input Voltage



$V_{GS} \leq V_t$

$v_O = v_{O2} - v_{O1}$

$i_{D2} = I - i_{D1}$

$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$

$v_{GS1} > v_{GS2}$

$v_{G1} - v_S > v_{G2} - v_S$

$\Rightarrow i_{D1} > i_{D2}$

$$I_D = I = \frac{1}{2} k_n' \frac{W}{L} (v_{GS1} - V_t)^2$$

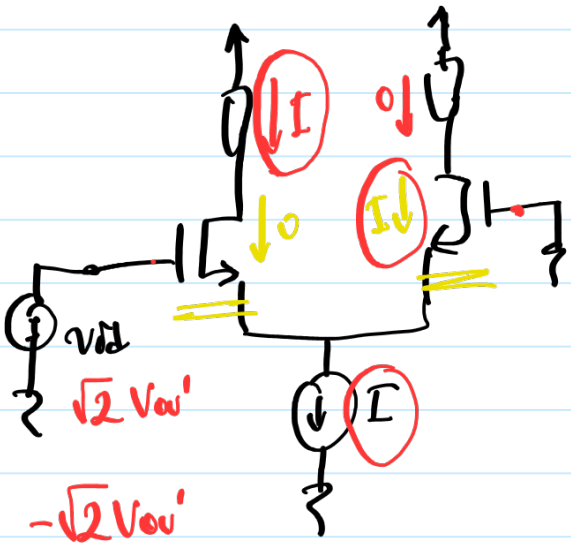
$$V_{GS} = V_t + \sqrt{\frac{2I}{k_n' W/L}} = V_t + \sqrt{2} \left(\frac{I}{k_n' W/L} \right)^{1/2}$$

$$V_{GS} = V_t + \sqrt{2}\, V_{ov}'$$

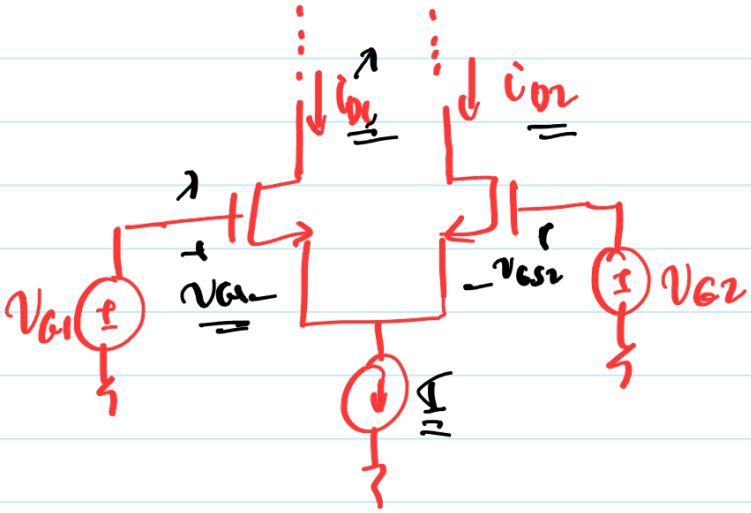
$$v_{id} = v_{GS1} + v_S = V_t + \sqrt{2}\, V_{ov}' - V_t$$

$$\boxed{v_{id} = \sqrt{2}\, V_{ov}'}$$

Large Signal Operation



$$-\sqrt{2}V_{ov}' \leq v_{id} \leq \sqrt{2}V_{ov}'$$



$$\sqrt{i_{D1}} = \sqrt{\frac{1}{2}k_n'\frac{W}{L}}\,(v_{GS1}-V_t)$$

$$\sqrt{i_{D2}} = \sqrt{\frac{1}{2}k_n'\frac{W}{L}}\,(v_{GS2}-V_t)$$

$$\left(\sqrt{i_{D1}}-\sqrt{i_{D2}}\right)^2 = \left(\sqrt{\frac{1}{2}k_n'\frac{W}{L}}\,\underbrace{(v_{GS1}-v_{GS2})}_{v_{id}}\right)^2$$

$$i_{D1}+i_{D2}=I \Rightarrow i_{D2}=I-i_{D1}$$

$$i_{D1}-2\sqrt{i_{D1}i_{D2}}+i_{D2}=\frac{1}{2}k_n'\frac{W}{L}v_{id}^2$$

$$I-\frac{1}{2}k_n'\frac{W}{L}v_{id}^2 = 2\sqrt{i_{D1}\underbrace{i_{D2}}_{I-i_{D1}}}$$

Large Signal Operation

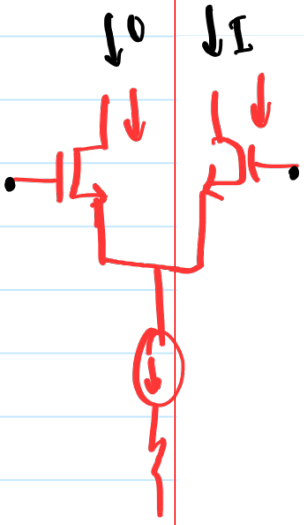
$$V_{ov}' = \frac{I}{k_n' W/L}$$

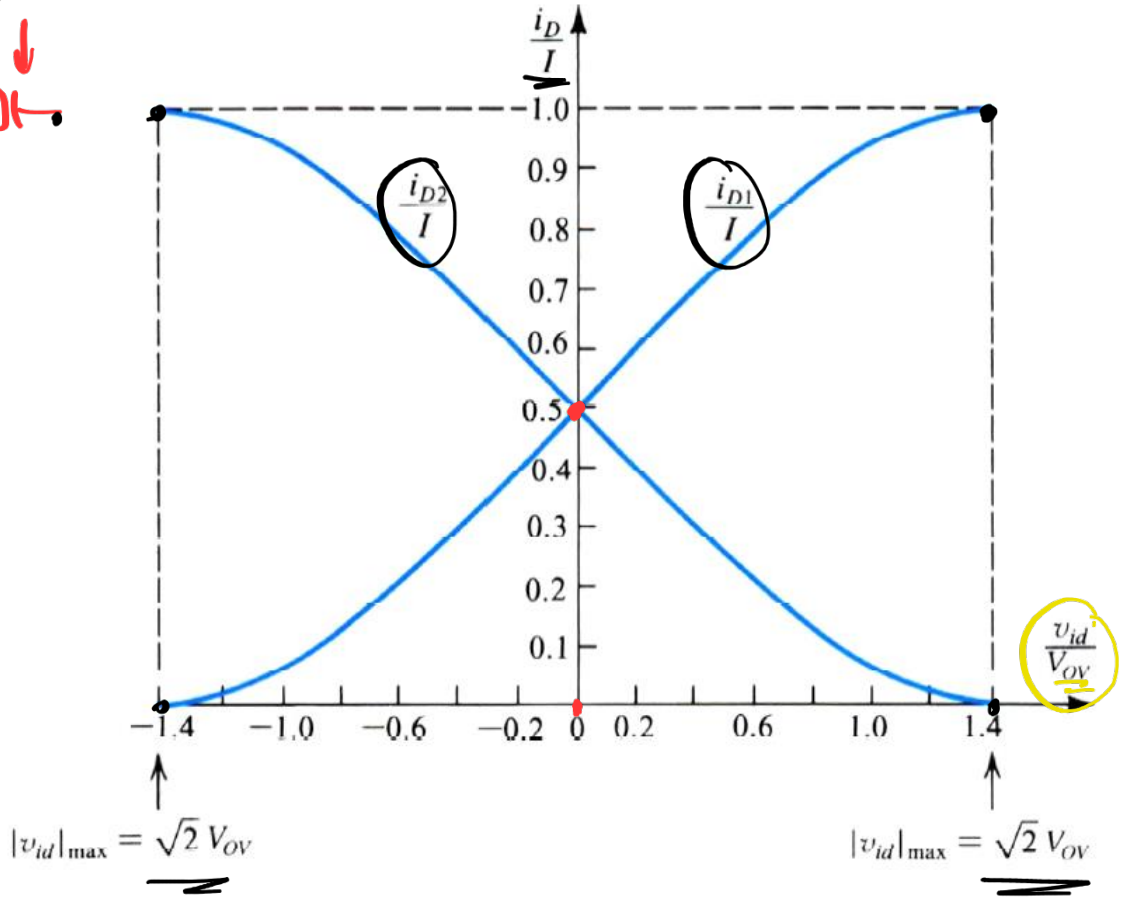
$$\left(I - \frac{1}{2} k_n' \frac{W}{L} v_{id}^2\right) = 4\, i_{D1}(I - i_{D1})$$

$$(\quad)^2 = 4 i_{D1} I - 4 i_{D1}^2$$

$$\frac{i_{D1}}{I} = \frac{1}{2} + \sqrt{k_n' \frac{W}{L}} \left(\frac{v_{id}}{2}\right) \sqrt{1 - \left(\frac{v_{id}/2}{V_{ov}'}\right)^2}$$

$$\frac{i_{D2}}{I} = \frac{1}{2} - \sqrt{k_n' \frac{W}{L}} \left(\frac{v_{id}}{2}\right) \sqrt{1 - \left(\frac{v_{id}/2}{V_{ov}'}\right)^2}$$





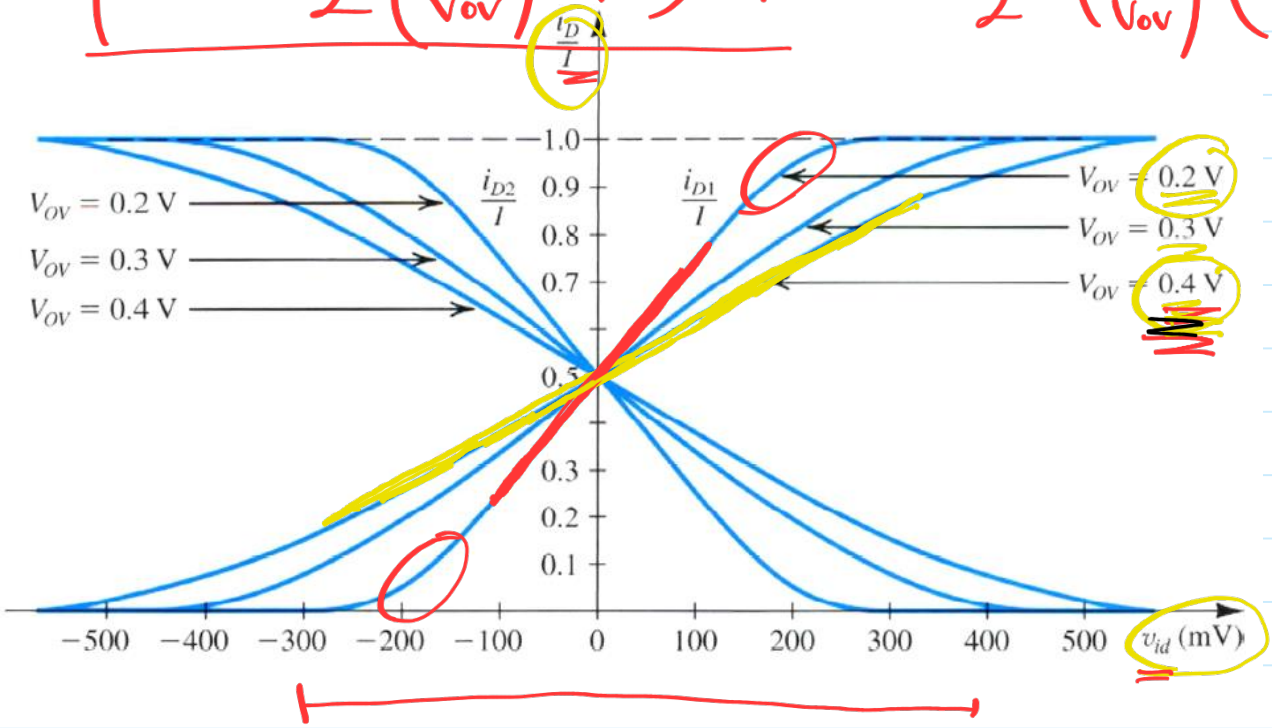
Large Signal Operation

$$\frac{i_{D1}}{I} = \frac{1}{2} + \sqrt{k_n' \frac{W}{L}} \left(\frac{v_{id}}{2}\right) \sqrt{1 - \frac{v_{id}/2}{V_{ov}'}}$$

$$i_{D1} = \frac{I}{2} + I\sqrt{k_n' \frac{W}{L}} \left(\frac{v_{id}}{2}\right)$$

$$v_{id} \ll 2V_{ov}'$$

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{ov}}\right)\cdot\left(\frac{v_{id}}{2}\right) \qquad i_{D2} = \frac{I}{2} - \left(\frac{I}{V_{ov}}\right)\cdot\left(\frac{v_{id}}{2}\right)$$



$$I_D = \frac{1}{2} k_n' \frac{W}{L} V_{ov}^2 \qquad g_m$$

Template